



PREDICTING EXTREME STOPPING TIME BEHAVIOR IN THE COLLATZ SYSTEM: A PROBABILISTIC AND STATISTICAL APPROACH

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ABSTRACT. The scope of this paper is the prediction of extreme stopping-time behaviour in the Collatz system, especially the statistical identification of integers whose normalized stopping time is unusually large. We study this problem from a probabilistic and statistical perspective. We first relate the phenomenon to the Collatz lattice, to a modified Collatz system with more regular convergence properties, and to the empirical density of long-orbit integers in finite windows. We then construct two logit models for extreme stopping-time behaviour using the modified-system stopping time and local spatial lags as predictors. The first model gives stable predictive performance near seventy percent across several scales, while the enhanced quadratic model improves both accuracy and specificity. We also introduce an interpretable decision-tree classifier for a moderately large stopping-time event, showing that lagged local information and normalized self-descent time contain substantial predictive signal. These results do not prove the Collatz conjecture, but they indicate that rare and moderately large stopping-time events have measurable local and dynamical structure that can be exploited statistically.

Predominant notation.

- $s(n)$: total stopping time of n under the original Collatz map.
- $sm(n)$: stopping time of n under the modified Collatz map.
- $Rs(n) = s(n)/\log_2(n)$: normalized, or relative, stopping time under the original Collatz map.
- $Rsm(n) = sm(n)/\log_2(n)$: normalized stopping time under the modified Collatz map.
- $Rs(n-1), Rs(n-2), Rs(n-3)$: local lag variables used in the logit models.
- $Y(n) > 7$, equivalently $Rs(n) > 7$: the extreme stopping-time event studied by the logit models.

1. **Introduction.** The Collatz conjecture, with its deceptively simple iteration:

$$\text{Col}(n) = \begin{cases} n/2, & \text{if } n \equiv 0 \pmod{2} \\ (3n+1)/2, & \text{if } n \equiv 1 \pmod{2} \end{cases}$$

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produces complex orbits whose behaviors remain unpredictable despite extensive study. At its heart lies the *stopping time* $s(n)$ —the number of iterations required for n to reach 1. While deterministic, the behavior of $s(n)$ exhibits features of chaos and pseudo-randomness, challenging direct mathematical analysis.

Iterating $\text{Col}(n)$ on any initial value $n \in \mathbb{N}$ produces a sequence known as the Collatz orbit:

$$n, \text{Col}(n), \text{Col}^2(n), \text{Col}^3(n), \dots \quad (1)$$

The Collatz conjecture asserts that every such sequence eventually reaches the value 1, regardless of the starting value. This behavior suggests an inherent convergence or stability property, and naturally invites interpretation as a discrete dynamical system:

$$x_{n+1} = \text{Col}(x_n), \quad x_0 = n. \quad (2)$$

An important indicator for the Collatz map is the stopping time $s(n)$ which is the smallest number of times the Collatz map must be applied to a number n in order to reach 1. Equivalently, the Collatz conjecture asserts that $s(n) < +\infty$ for every natural number n .

Despite its elementary formulation, the Collatz conjecture has resisted proof for decades and remains open. Numerous studies ([1–3]) have yielded valuable partial results. For instance, Terras (1976) [4] demonstrated that almost all initial values (interpreted as “density one” or heuristically as “more than 99.99%”) eventually descend below their starting point. This foundational result was later refined: Allouche (1979) [5] showed that almost all orbits reach a value less than $n^{0.869}$, while Korec (1994) [6] improved this bound to $n^{0.7925}$.

A potential path to disproving the conjecture would be the existence of a non-trivial cycle, distinct from the known loop $1 \rightarrow 2 \rightarrow 1$. However, it has been rigorously shown [7] that any such cycle must contain at least 17,087,915 elements—making it computationally infeasible to discover a short counterexample.

In 2003, Krasikov and Lagarias showed [5] (with a computer-assisted proof) that for any large number x , there were at least $x^{0.84}$ initial values n between 1 and x whose Collatz iteration reached 1. In 2017, a distributed computing project verified the Collatz conjecture for all starting values n up to 10^{20} . So it is highly unlikely that a counterexample can be found just from pen and paper search. In 2019, Tao showed [8] that almost all initial values n eventually iterated to a value less than $f(n)$, for any function $f(n)$ that grew to infinity, no matter how slowly. Therefore almost all Collatz orbits attain almost bounded values. For instance, almost all initial values n eventually iterate to a value less than $\log(\log(\log(\log(n))))$. This is about as close as one can get to the Collatz conjecture without actually solving it. The argument was inspired by other dynamical systems results, and in particular by a 1994 result of Bourgain [2] on constructing an invariant measure for the nonlinear Schrödinger equation.

The present paper is placed in this probabilistic tradition, but it addresses a different problem from the classical density theorems. Terras [9] used parity vector arguments and the central limit theorem to show that, for almost all initial values, the orbit eventually descends below its starting point. Allouche [1] and Korec [4] refined this line of work by proving stronger descent estimates, with Korec showing that almost all orbits descend below $n^{0.7925}$. Krasikov and Lagarias [5] obtained quantitative lower bounds for the number of integers below x whose iterates reach 1, while Lagarias’ surveys [6, 7] emphasise the mixture of arithmetic structure, stochastic behaviour, residue class effects, and distributional phenomena, including

Benford-type behaviour, that appear in Collatz dynamics. Tao’s theorem [8] is the strongest result in this hierarchy: it shows, in logarithmic density, that almost all orbits attain values below any prescribed function tending to infinity, using deep 3-adic Fourier analysis and renewal type ideas. These results describe the rarity of exceptional orbits. By contrast, the present paper asks a predictive question: given an explicitly sampled integer n , can one identify whether it belongs to the extreme stopping time tail $Rs(n) > 7$ without computing the full orbit? In this sense, the logit models developed below do not compete with the density one results of Terras, Allouche, Korec, Krasikov–Lagarias, and Tao; rather, they seek to locate and statistically characterise the exceptional set whose small density those theorems reveal.

This distinction is also reflected in the variables used in our models. The condition $Rs(n) > 7$ selects orbits whose total stopping time is far larger than the natural logarithmic scale of n and hence corresponds to an anomalous imbalance in the parity sequence. The modified Collatz relative stopping time $Rsm(n)$ acts as a computable surrogate for this imbalance in a system where convergence is controlled and the heavy tail is suppressed, while the spatial lag variables $Rs(n-1)$, $Rs(n-2)$, $Rs(n-3)$ record local clustering in the Collatz lattice. Thus the methodology developed here does not prove a new density theorem and does not exclude possible exceptional cases left open by the known almost-all results. Instead, it opens an empirical window into where rare extreme stopping-time events occur, using logit regression to predict the event $Rs(n) > 7$ with approximately 70%–75% accuracy across both training and out-of-sample ranges. Misclassified cases and residual patterns may therefore be interpreted as candidates for further arithmetic investigation, including possible residue class or Benford-type structure in the exceptional tail.

Statistics plays a crucial role in the analysis and understanding of dynamical systems. Statistical methods are used to estimate parameters, select models, reduce dimensionality, quantify noise and uncertainty, and detect chaos. Dynamical systems often involve parameters that govern their behavior. Statistical methods are used to estimate these parameters from observed data. This is essential for model validation, prediction, and control.

In recent years, there has been growing interest in applying statistical methods to study the Collatz Conjecture. By analyzing the statistical properties of the Collatz sequence, researchers hope to gain new insights into its behavior and potentially uncover hidden patterns. Statistical methods offer a powerful tool for analyzing large datasets and identifying trends that may not be apparent through traditional mathematical techniques.

This work concentrates on a data-driven approach to large stopping-time behaviour. We construct two logit models predicting the occurrence of *extreme stopping times*, specifically when the normalized value $Rs(n) = s(n)/\log_2(n) \geq 7$, and we also develop an interpretable decision-tree classifier for the moderately large event $Y(n) > 5$. These approaches bypass full Collatz computation for the target integer by using *surrogate variables* from a simplified Collatz-like system, spatial lag patterns, residue-class indicators, and self-descent information.

The use of the normalized, or relative, stopping time rather than the unnormalised stopping time is important because the typical scale of $s(n)$ grows roughly with $\log_2(n)$. Thus $Rs(n) = s(n)/\log_2(n)$ separates genuinely exceptional orbit behaviour from the trivial increase in stopping time caused merely by starting from

a larger integer. The threshold $Rs(n) > 7$ therefore identifies integers whose orbits are unusually long relative to their natural logarithmic scale, making it a more stable target for comparison across wide numerical ranges.

This paper is organised as follows. Section 2 introduces the Collatz lattice and the logarithmic identity that motivates the geometric viewpoint. Section 3 defines the modified Collatz system and compares its relative stopping-time distribution with that of the original system. Section 4 studies the empirical density of long-orbit integers satisfying $Y(n) > 7$, using fixed windows and a log-uniform Monte Carlo experiment, and documents the strong clustering of the dependent variable. Section 5 presents Model 1, the four-variable logit model, together with its estimation, training performance, and multi-scale validation. Section 6 develops Model 2, the enhanced eight-variable quadratic logit model, and compares it with Model 1. Section 7 introduces the decision-tree classifier for the moderately large event $Y(n) > 5$, reports its test performance, and interprets its main rules and feature importances. Section 8 concludes.

2. Mathematical context and the collatz lattice. The complex behaviour of the Collatz map is manifested even in its application at small natural numbers. From the orbits displayed in Figure 1 it becomes apparent that numbers well separated, like 170 and 512, or 53 and 160, have the same stopping time $s(n)$.

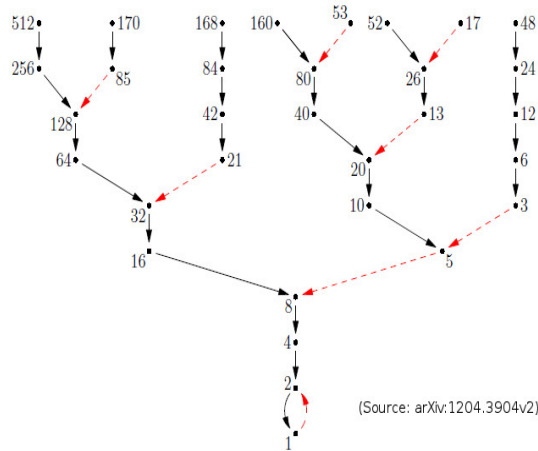
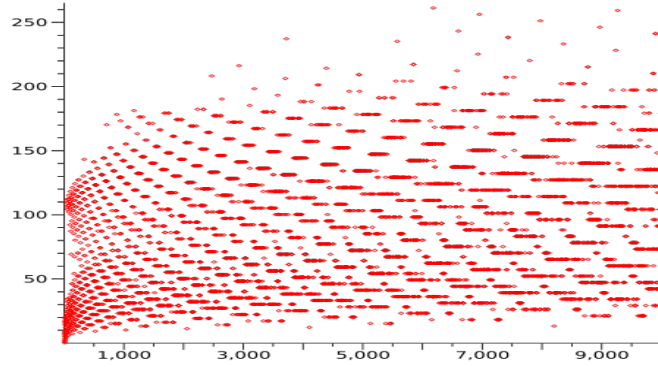
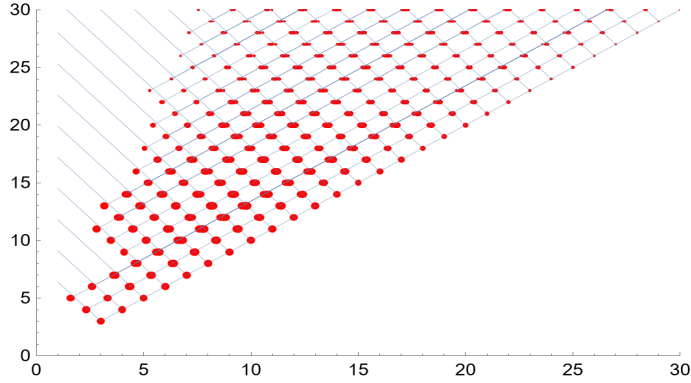


FIGURE 1. The vertices of the directed graph, called Collatz tree, are occupied by the natural numbers. The arrows in the graph show the results of the application of the Collatz map.

This complex behaviour is demonstrated on a larger scale in Figure 2 where the total stopping time $s(n)$ of numbers from 1 to 9999 is portrayed. Figure 2 portrays also the chaotic characteristics of the Collatz dynamics: nearby natural numbers may have totally different stopping times and therefore they may belong to very different Collatz orbits.

However, in this apparently chaotic and complex behaviour of the Collatz dynamics there is an implicit order which becomes manifest if in Figure 2 we replace n in the horizontal axis by $\log_2(n)$. This is demonstrated in Figure 3.


 FIGURE 2. The stopping time $s(n)$ against n .

 FIGURE 3. The stopping time $s(n)$ against $\text{Log}_2(n)$. This is the level set of the Collatz map. It becomes evident that the Collatz map distributes the natural numbers in a lattice type of graph.

The natural numbers arrange themselves on the lattice in Figure 3 according to Equation (3):

$$\text{Log}_2(n) = s(n) - \text{Log}_2(3)d(n) - \sum_i^{d(n)} \text{Log}_2\left(1 + \frac{1}{3o_i}\right). \quad (3)$$

For completeness, we recall the derivation of this identity. During an odd step starting from an odd value o_i , the accelerated Collatz map multiplies the current value by $3(1 + 1/(3o_i))/2$, while every even step contributes a factor $1/2$. If the orbit reaches 1 after $s(n)$ total steps and contains $d(n)$ odd steps, then

$$1 = n \frac{3^{d(n)} \prod_{i=1}^{d(n)} \left(1 + \frac{1}{3o_i}\right)}{2^{s(n)}}.$$

Taking base-two logarithms gives (3). This is the standard logarithmic bookkeeping identity for accelerated Collatz orbits; see, for example, Lagarias' overview of the $3x + 1$ problem. The first two terms on the right hand side of Equation (3) generate the motion on the lattice in Figure 3. The term $s(n)$ causes the displacement in the vertical direction whereas the term $\text{Log}_2(3)d(n)$ causes the displacement in

the horizontal direction. The small number $\sum_i^{d(n)} \text{Log}_2 \left(1 + \frac{1}{3\sigma_i}\right)$ causes the motion in the neighborhood of each vertex of the lattice creating the small clusters in the vicinity of each vertex of the lattice. The identity (3) suggests that $\text{Log}_2(n)$ lies on near-linear trajectories determined by stopping time statistics. This leads to a *natural coordinate system* for the Collatz dynamics, allowing statistical clustering and regression modeling.

In order to illustrate the meaning of Equation (3) take $n = 10$. The associated Collatz orbit is

$$10 \longrightarrow 5 \longrightarrow 8 \longrightarrow 4 \longrightarrow 2 \longrightarrow 1.$$

Therefore, $s(10) = 5$ and $d(10) = 1$. Equation (3) reads

$$s(10) - \text{Log}_2(3)d(10) - \text{Log}_2 \left(1 + \frac{1}{3 \cdot 5}\right) = \text{Log}_2 \left(\frac{5 \cdot 32}{16}\right) = \text{Log}_2(10). \quad (4)$$

Given that $\text{Log}_2(n) \leq s(n)$ and that $\text{Log}_2(3)$ is the average distance between adjacent clusters with the same $s(n)$ Equation (4) places $\text{Log}_2(10)$ in Figure 3 on the line $s(n) = 5$ in the second cluster from the right.

3. The modified collatz system and distribution of relative stopping times.

To decouple the analysis from the original conjecture, we construct a *modified Collatz map*:

$$\text{Col}_{\text{mod}}(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2} \\ (3n+1)/2, & n \equiv 1 \pmod{4} \\ (3n-1)/2, & n \equiv 3 \pmod{4} \end{cases}$$

This map preserves the fixed point $\{1, 2\}$ but has vastly simplified convergence properties. The use of modified or accelerated versions of the $3x+1$ map is common in the literature on stopping times and stochastic models of Collatz-type dynamics; see, among others, Terras [9], Lagarias [6, 7], and Wirsching [10]. Here $sm(n)$ denotes the stopping time of n under the modified map Col_{mod} , i.e. the least number of modified iterations required to reach the cycle $\{1, 2\}$. The stopping times $sm(n)$ of this system exhibit:

- Narrower variance than $s(n)$,
- Similar cluster structure under the $\log_2(n)$ transformation,
- No outliers like the classic $n = 27$ in Collatz.

These properties make $sm(n)$ a robust *explanatory proxy* for $s(n)$ in statistical models.

Let us denote by T_1, T_2 the Collatz and the modified Collatz dynamic systems respectively. For convenience we define the notion of Level set graph:

Level set graph : The graph $LT_i = \{(\text{Log}_2(n), s(n)), n \in N\}$ where $s(n)$ is the stopping time of n for the dynamical system defined by T_i , $i = 1, 2$. For example the graph in Figure 3 is the level set graph LT_1 for the dynamical system defined by T_1 .

The level set graphs LT_1 and LT_2 given up to level 30, for the dynamical systems defined by T_1 and T_2 are compared in Figure 4.

Level set graphs LT_1 and LT_2 have similar lattice structure with additive common ratios to be $(\text{Log}_2(3), 1)$ for the two dimensions. Every level set at height “ s ” has as range the interval $w\text{Log}_2(s) + c, s$ (where w, c are suitably defined) and its density is concentrated at the lattice points $s - k\text{Log}_2(3)$ where $k \in (0, (s - (w\text{Log}_2(s) + c))/\text{Log}_2(3)) \cap N$.

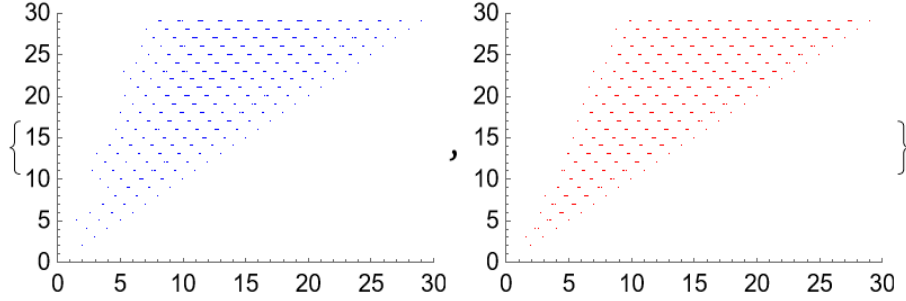


FIGURE 4. The stopping time $s(n)$ against $\text{Log}_2(n)$ for the Collatz and the modified Collatz maps. These are the level set graphs LT_1 and LT_2 for the Collatz and the modified Collatz systems respectively.

Although the graphs in Figure 4 look very similar their level sets LT_1 and LT_2 differ in cardinality and rate of growth. In fact the level set LT_2 of modified Collatz is much larger and with higher growth rate than the level set LT_1 of T_1 . The evolution of the logarithm of the cardinality of the level sets of the modified (yellow) and the original (blue) Collatz is depicted in Figure 5.

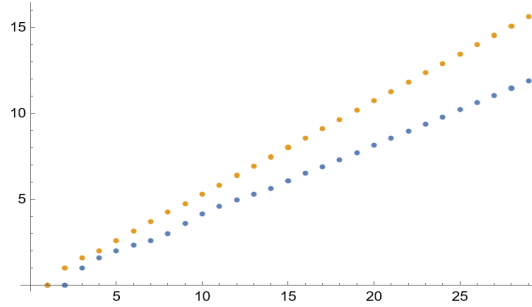


FIGURE 5. The Logarithm of the cardinality of the level sets LT_1 and LT_2 against the stopping time $s(n)$.

In Figure 5 the logarithm of the cardinality of the level sets LT_1 and LT_2 is displayed on the horizontal axis whereas the stopping time is displayed on the vertical axis. Figure 5 shows the growth of level sets for the Collatz (blue) and the modified Collatz (yellow) systems.

3.1. Distribution of relative stopping times. We define the relative stopping times $Rs(n) = \frac{s(n)}{\text{Log}_2(n)}$ and $Rsm(n) = \frac{sm(n)}{\text{Log}_2(n)}$ as follows:

$$Rs(n) = \frac{s(n)}{\text{Log}_2(n)}, \quad Rsm(n) = \frac{sm(n)}{\text{Log}_2(n)}. \quad (5)$$

We calculated the distribution of the relative stopping times for the two collatz dynamical systems on 4000 numbers randomly selected in $[10^4, 10^7]$.

It is apparent based on the study of this sample that $Rsm(n)$ is significantly smaller to $Rs(n)$ and with much smaller variation. Also it does not exhibit numbers with disproportionate stopping times.

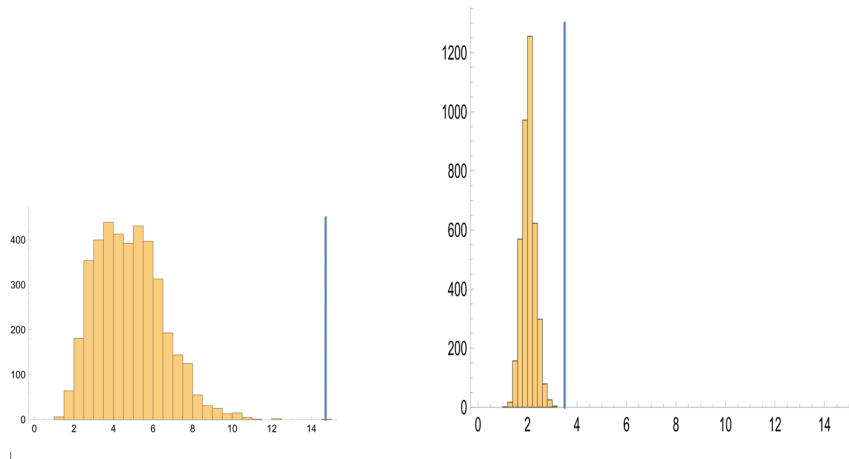


FIGURE 6. Distribution of $Rs(n)$, on the left hand side, and of $Rsm(n)$ on the right hand side of 4000 numbers randomly selected in $[10^4, 10^7]$. On the horizontal axis $Rs(n)$ is given on the left hand side, and of $Rsm(n)$ on the right hand side. On the vertical axis the absolute frequency is displayed. The sample distribution of $Rs(n)$ exhibits a heavy tail bounded from the right by $Rs(27)$ (blue line), and the sample distribution of $Rsm(n)$ shows a tight bell-shaped distribution bounded from the right by $Rs(12233)$ (blue line).

4. Density of long-orbit integers in the collatz system: Estimating the proportion of n with $Y(n) > 7$. The dependent variable in the logit models is the indicator of the event $Y(n) > 7$, equivalently $Rs(n) > 7$ in the notation used elsewhere in the paper. Before using this event as the object of prediction, it is necessary to examine how frequently it occurs, how strongly its frequency depends on the location of the interval sampled, and whether the exceptional long-orbit integers are distributed uniformly or instead appear in local clusters.

We study the proportion of integers n in a finite interval S whose normalized total stopping time

$$Y(n) = \frac{s(n)}{\log_2(n)}$$

exceeds the threshold 7, where $s(n)$ is the number of Collatz iterations required for the orbit to reach the terminal cycle. The threshold $Y(n) > 7$ identifies integers whose Collatz orbit is more than seven times longer than their bit-length, and is therefore a strong form of the long-orbit property. This variable is the main object of the present statistical study; it constitutes the dependent variable of the models considered below, and hence the exploration of its density is of central importance for the foundation of the paper.

Formally, for a finite window S , we seek to estimate

$$p(S) = \frac{|\{n \in S : Y(n) > 7\}|}{|S|}, \quad (6)$$

for windows of the form $S = [a, a + 10^4]$, each of length 10,000. The analysis proceeds in three steps: first, we compute $p(S)$ for six specific windows anchored

at powers of 10 and powers of 3, respectively; second, we conduct a Monte Carlo study drawing $K = 200$ windows uniformly at random on the logarithmic scale from $[2, 10^{40}]$; and third, we discuss what these results reveal about the distribution of long-orbit integers within $\mathbb{N}$.

4.1. Specific window estimates.

4.1.1. *Windows anchored at powers of 10.* Figure 7 reproduces the three equal-length windows anchored at powers of 10. These are the windows used for direct comparison with the samples considered earlier in the paper. The three estimates are 0.1059, 0.0937, and 0.0051, respectively, showing that the local density of the long-orbit event may fall sharply even between windows of the same length.

Sample	Window $S = [a, a+10^4]$	Count $\delta=1$	$\hat{p}(S)$	95% CI	Mean $Y(n)$	Std $Y(n)$
Ssm	$[10^6, 10^6 + 10^4]$	1,059 / 10,000	0.1059	[0.0999, 0.1119]	4.594	1.766
Smed	$[10^8, 10^8 + 10^4]$	937 / 10,000	0.0937	[0.0880, 0.0994]	4.238	1.540
Slar	$[10^{15}, 10^{15} + 10^4]$	51 / 10,000	0.0051	[0.0037, 0.0065]	5.320	0.565

FIGURE 7. Windows anchored at powers of 10. The diagram reports, for $S_{sm} = [10^6, 10^6 + 10^4]$, $S_{med} = [10^8, 10^8 + 10^4]$, and $S_{lar} = [10^{15}, 10^{15} + 10^4]$, the count of integers satisfying $Y(n) > 7$, the estimated proportion $\hat{p}(S)$, a 95% confidence interval, and the mean and standard deviation of $Y(n)$ within the window. It shows that the estimate declines from 10.59% and 9.37% in the two smaller power-of-10 windows to only 0.51% at 10^{15} , where the standard deviation of $Y(n)$ is also very small.

4.1.2. *Windows anchored at powers of 3.* Figure 8 presents the same computation for three windows whose lower endpoint is a power of 3. Powers of 3 are natural reference points in the Collatz system: the inverse Collatz map has denominators 3^d , so neighbourhoods of 3^k are structurally significant. The starting integers are

$$Sm : 3^{13} = 1,594,323 \quad (\log_2 \approx 20.6),$$

$$Med : 3^{18} = 387,420,489 \quad (\log_2 \approx 28.5),$$

$$Larg : 3^{30} = 205,891,132,094,649 \quad (\log_2 \approx 47.5).$$

The corresponding estimated proportions are 0.1692, 0.0531, and 0.1145. Thus, unlike the power-of-10 windows, the largest power-of-3 window is not a low-density region; it lies in a comparatively high-density cluster of long-orbit integers.

Sample	Window S	Count $\delta=1$	$\hat{p}(S)$	95% CI	Mean $Y(n)$	Std $Y(n)$	$\log_2(a)$
Sm	$[3^{13}, 3^{13}+10^4]$	1,692 / 10,000	0.1692	[0.1619, 0.1765]	4.799	1.927	20.61
Med	$[3^{18}, 3^{18}+10^4]$	531 / 10,000	0.0531	[0.0487, 0.0575]	4.719	1.388	28.53
Larg	$[3^{30}, 3^{30}+10^4]$	1,145 / 10,000	0.1145	[0.1083, 0.1207]	4.979	1.206	47.55

FIGURE 8. Windows anchored at powers of 3. The diagram gives the same statistics as Figure 7, now for the intervals $[3^{13}, 3^{13} + 10^4]$, $[3^{18}, 3^{18} + 10^4]$, and $[3^{30}, 3^{30} + 10^4]$, with the additional column $\log_2(a)$. It shows that the local density need not decrease monotonically with scale: the estimate falls from 16.92% at 3^{13} to 5.31% at 3^{18} , but rises again to 11.45% at 3^{30} .

4.1.3. *Comparison and first observations.* The six windows span an enormous range, from 10^6 to $3^{30} \approx 2 \times 10^{14}$, yet the estimates do not decrease monotonically. Across Figures 7 and 8, the values of $\hat{p}(S)$ are

$$S_{sm} : 10.6\%, \quad S_{med} : 9.4\%, \quad S_{lar} : 0.5\%,$$

$$Sm : 16.9\%, \quad Med : 5.3\%, \quad Larg : 11.5\%.$$

Two striking patterns emerge. First, a broadly decreasing trend with magnitude is visible when comparing $S_{sm} \rightarrow S_{med}$, where the lower endpoint moves from 10^6 to 10^8 and the proportion falls from 10.6% to 9.4%, and when comparing $Sm \rightarrow Med$, where the lower endpoint moves from 3^{13} to 3^{18} and the proportion falls from 16.9% to 5.3%. This is consistent with the theoretical concentration of $Y(n)$ around its mean as $n \rightarrow \infty$, which shrinks the tail $Y(n) > 7$.

Second, and more importantly, the decreasing trend is not monotone. The window *Larg*, at $3^{30} \approx 2 \times 10^{14}$, gives 11.5%, far above S_{lar} , at 10^{15} , which gives only 0.5%, despite the two windows lying at comparable large scales. This non-monotone behaviour reflects the local clustering of long-orbit integers: the window starting at 3^{30} falls inside a high-density cluster, whereas the window starting exactly at 10^{15} falls inside a short-orbit cluster. The standard deviation of $Y(n)$ within each window provides further evidence of this clustering. The window S_{lar} has $\sigma(Y) = 0.565$, an exceptionally narrow distribution in which nearly all integers have $Y \approx 5.3$, confirming that it sits in a homogeneous short-orbit neighbourhood. The other windows have $\sigma(Y) > 1.2$, reflecting the typical heterogeneity of Y across a generic interval.

4.2. Monte carlo estimation.

4.2.1. *Methodology.* To obtain a scale-neutral summary estimate, we draw $K = 200$ windows $S = [a, a + 10^4]$, where the starting point a is drawn log-uniformly from $[2, 10^{40}]$, meaning

$$\log_{10}(a) \sim \text{Uniform}[0, 40].$$

Log-uniform sampling ensures equal representation of each order of magnitude, rather than linear sampling, which would overwhelmingly favour the largest values.

For each window we compute $\hat{p}(S)$ exactly by tracing every Collatz orbit. The 200 windows are distributed across five equal magnitude bands:

$$[2, 10^8), \quad [10^8, 10^{16}), \quad [10^{16}, 10^{24}), \quad [10^{24}, 10^{32}), \quad [10^{32}, 10^{40}).$$

4.2.2. *Results.* Figure 9 reports the Monte Carlo estimates for the five magnitude bands and for the overall sample of $K = 200$ windows. The band means decrease from 0.1678 in the smallest band to 0.0204–0.0244 in the two largest bands, while the observed maximum remains very large even at high scales, reaching 0.3604 and 0.3631 in the two largest bands. The overall log-uniform mean is 0.0606, with standard deviation 0.0830, minimum 0, and maximum 0.3631.

Magnitude band	K windows	Mean $\hat{p}$	Std dev	Min	Max
$\sim 10^4$ $[2, 10^8)$	40	0.1678	0.0568	0.0288	0.2276
$\sim 10^{12}$ $[10^8, 10^{16})$	40	0.0598	0.0557	0.0034	0.2413
$\sim 10^{20}$ $[10^{16}, 10^{24})$	40	0.0307	0.0581	0.0004	0.2684
$\sim 10^{28}$ $[10^{24}, 10^{32})$	40	0.0204	0.0593	0.0000	0.3604
$\sim 10^{36}$ $[10^{32}, 10^{40})$	40	0.0244	0.0767	0.0000	0.3631
Overall (K=200)	200	0.0606	0.0830	0.0000	0.3631

FIGURE 9. Monte Carlo estimates of $\hat{p}(S)$ for $K = 200$ windows of length 10,000, with starting points sampled log-uniformly from $[2, 10^{40}]$. The diagram reports, by magnitude band, the number of sampled windows, the mean of $\hat{p}(S)$, its standard deviation, and the minimum and maximum observed local proportions. It shows both scale dependence, through the decline in the band means, and strong non-uniformity, through the wide min–max ranges that persist even for windows at magnitudes 10^{24} to 10^{40} .

4.3. Discussion: non-uniform distribution of long-orbit integers.

4.3.1. *The clustering phenomenon.* The most important finding is that the set

$$\{n \in \mathbb{N} : Y(n) > 7\}$$

is not uniformly distributed within $\mathbb{N}$. Three lines of evidence converge on this conclusion. First, the standard deviation of $\hat{p}$ across the 200 Monte Carlo windows is 0.083, which exceeds the mean 0.061. The coefficient of variation is therefore

$$CV = \frac{0.083}{0.061} \approx 1.37 > 1,$$

so the distribution is more variable than the Poisson regime, a hallmark of clustering. Second, the median $\hat{p} = 0.017$ is less than one-third of the mean 0.061. This extreme skew indicates that the mean is dominated by a small number of very high-density windows: half of all windows have fewer than 1.7% of integers with $Y > 7$, while a small fraction have more than 20%. Third, the six specific-window estimates vary from 0.51% to 16.92%, a 33-fold range across windows of identical size, confirming that the local density is highly location-dependent.

4.3.2. Scale dependence. There is also a genuine scale effect overlaying the clustering. The band-by-band means decrease from 16.8%, near 10^4 , to 2.0%–2.4%, near 10^{32} – 10^{40} . This reflects the theoretical concentration of $Y(n)$ near its mean of approximately 4.8 as $n \rightarrow \infty$. As the standard deviation of Y contracts with scale, the tail $Y > 7$, located roughly 1.8–2.2 standard deviations above the mean, becomes progressively thinner. However, this trend is masked locally by clustering, as demonstrated by the 11.5% estimate at 3^{30} , which is far larger than the 2%–3% band average near that scale.

4.3.3. An estimate of the proportion. Given the above, the natural question is what single estimate, if any, should be assigned to the proportion

$$p = \frac{|\{n \in [1, N] : Y(n) > 7\}|}{N}$$

for large N . A single fixed density does not exist in a practically useful finite-window sense. The proportion is neither a well-defined constant at these scales nor a simple function of $\log N$. Instead, for N in the range 10^6 – 10^8 , the local proportion is approximately 9%–17%, dominated by cluster effects. For very large N , in the range 10^{20} – 10^{40} , the band-average proportion falls to 2%–3%, but individual windows vary from 0% to over 36%. The log-uniform Monte Carlo mean of 6.1%, with 95% confidence interval [4.9%, 7.2%], provides a scale-averaged summary estimate, but the median of 1.7% is arguably more representative of a generic window.

The conclusion is that the density of $\{n \in \mathbb{N} : Y(n) > 7\}$ is not captured by a fixed proportion. It is highly variable across windows of a fixed length, ranging from 0% to over 36% in windows of 10,000 consecutive integers. The variability itself, measured by a coefficient of variation exceeding 1.3, is the primary statistical descriptor of the set. A range of [0%, 36%], with a log-uniform mean of approximately 6% and 95% confidence interval [4.9%, 7.2%], constitutes the most honest summary estimate. The decreasing trend with magnitude, from approximately 17% near 10^4 to approximately 2% near 10^{36} , is real but locally overwhelmed by clustering. This is precisely why a local predictive framework, using variables such as $Rs(n-1)$, $Rs(n-2)$, $Rs(n-3)$, and the modified-system quantity $Rsm(n)$, is appropriate for the logit analysis that follows.

5. Model 1: The logit model. specification, estimation, and multi-scale validation.

5.1. Motivation and model specification. It is important to understand why some numbers have large relative stopping time $Rs(n) = s(n)/\log_2(n)$. To this end we model the probability that $Rs(n) \geq 7$ using four explanatory variables: $Rsm(n)$, the relative stopping time of n under the modified Collatz system T_2 defined in Section 3, and the lagged relative stopping times $Rs(n-1)$, $Rs(n-2)$, $Rs(n-3)$ of the three immediate predecessors of n .

The choice of $Rsm(n)$ as the primary continuous predictor is motivated by the structural proximity of T_2 to the original Collatz map: T_2 is a small perturbation of T_1 whose convergence is rigorously understood. Despite having lower absolute values and smaller variance than $Rs(n)$, $Rsm(n)$ carries genuine information about the orbit length of n . The lagged terms $Rs(n-k)$, for $k = 1, 2, 3$, capture the well-documented empirical clustering of long-orbit integers in the integer line: numbers with large stopping times tend to appear in consecutive runs, a consequence of the inverse Collatz tree structure.

The logit model takes the form

$$P(Rs(n) \geq 7) = \frac{1}{1 + \exp[-\eta_1(n)]}, \quad (7)$$

$$\eta_1(n) = c + \alpha_1 Rsm(n) + \alpha_2 Rs(n-1) + \alpha_3 Rs(n-2) + \alpha_4 Rs(n-3).$$

where $c, \alpha_1, \alpha_2, \alpha_3, \alpha_4$ are real parameters to be estimated by maximum likelihood.

5.2. Training sample design and statistical power. We address the question of sample size and event-per-variable requirements by adopting a balanced sampling design. Specifically, the training sample Tr consists of 1,536 integers drawn from $[10^4, 10^9]$, comprising exactly 768 integers with $Rs(n) \geq 7$, the positive class, and 768 integers with $Rs(n) < 7$, the negative class. The balanced design serves three purposes.

First, it ensures event-per-variable compliance. With 768 events per class and four predictors, the effective event-per-variable ratio is

$$\frac{768}{4} = 192,$$

far exceeding the commonly cited threshold of 10, and providing ample power for stable coefficient estimation. Second, it gives robustness to class imbalance. The overall base rate of $Rs(n) > 7$ varies strongly with the magnitude of n and with the specific window of integers sampled, ranging from near 0% in short-orbit clusters to over 35% in long-orbit clusters, as established in the preceding analysis of the density of long-orbit integers. A balanced design decouples classification performance from this scale-dependent base rate. Third, it ensures comparability across scales: balanced test sets at each validation scale make the reported accuracy figures reflect genuine discriminative power, rather than the performance of a trivial majority-class classifier.

We draw balanced samples of integers by iterating through uniform draws from the target range and retaining each draw in the appropriate class pool until both pools reach the target size. The final training set is randomly shuffled prior to estimation.

5.3. Estimation results. Maximum likelihood estimation of model (7) on the balanced training sample Tr yields the coefficient estimates reported in Table 1. Standard errors are obtained by the bootstrap with 200 resamples. All coefficients are positive, consistent with the intuition that larger relative stopping times in the modified system and in the neighbourhood of n increase the probability of $Rs(n) \geq 7$.

Therefore, the estimated logit model reads

$$P(Rs(n) \geq 7) = \frac{1}{1 + \exp(\eta_2(n))}, \quad (8)$$

$$\eta_2(n) = 4.3646 - 0.4913 Rsm(n) - 0.3627 Rs(n-1) - 0.2021 Rs(n-2) - 0.0694 Rs(n-3).$$

The strongest predictor, by both coefficient magnitude and permutation importance, is $Rs(n-1)$, the relative stopping time of the immediate predecessor. Permuting $Rs(n-1)$ reduces training accuracy by 11.4 percentage points, compared to 5.1 percentage points for $Rs(n-2)$, 0.7 percentage points for $Rsm(n)$, and 0.4 percentage points for $Rs(n-3)$. This ranking reflects the strong autocorrelation of orbit lengths across consecutive integers, which is in turn explained by the inverse

Variable	Estimate	Std. Error	z-Statistic	p-value
Intercept (c)	-4.3646	0.5672	-7.695	$< 10^{-14}$
$x_1 = Rsm(n)$	0.4913	0.2304	2.132	0.0330
$x_2 = Rs(n-1)$	0.3627	0.0360	10.070	$< 10^{-23}$
$x_3 = Rs(n-2)$	0.2021	0.0392	5.154	$< 10^{-6}$
$x_4 = Rs(n-3)$	0.0694	0.0366	1.897	0.0578

TABLE 1. Coefficient estimates for logit model (7), estimated on the balanced training sample of 1,536 integers from $[10^4, 10^9]$. Bootstrap standard errors are computed using 200 resamples. The table shows that all estimated effects are positive: large values of the modified-system stopping time and of the neighbouring stopping times raise the estimated probability of the long-orbit event $Rs(n) \geq 7$.

Collatz tree structure: integers in a neighbourhood of n often share inverse-tree ancestors and therefore tend to have similar stopping times. The modified-system predictor $Rsm(n)$ is the strongest among individual structural predictors of n itself, confirming that the modified Collatz dynamics serve as a robust proxy for the original system.

Table 2 reports the standard information-theoretic fit statistics.

Statistic	Value
Log-Likelihood	-924.90
Adjusted McFadden Pseudo R^2	0.1266
AIC	1,859.80
BIC	1,886.49

TABLE 2. Goodness-of-fit statistics for logit model (8), balanced training sample ($n = 1,536$). The adjusted McFadden pseudo R^2 value indicates acceptable fit for a binary logit model, while AIC and BIC provide information criteria for comparison with the enhanced model considered later.

The adjusted McFadden R^2 of 0.127 is consistent with the range typically regarded as indicating acceptable fit in binary logit models; values of 0.10–0.30 are often interpreted as analogous to substantially larger R^2 values in linear regression. The AIC and BIC are used for model comparison with the enhanced specification.

5.4. Training set performance. The confusion matrix on the training set, using the cut-off 0.5, is given in Table 3.

The overall training accuracy is 69.3%, with sensitivity, that is recall of the positive class $Rs(n) \geq 7$, equal to 67.3%, and specificity equal to 71.4%. The near-symmetry between sensitivity and specificity reflects the balanced design. The training accuracy of 69.3% establishes the in-sample benchmark for the model’s discriminative capacity, against which out-of-sample performance is assessed in the next subsection.

	Predicted $\delta = 0$	Predicted $\delta = 1$
Actual $\delta = 0$	548 (TN)	220 (FP)
Actual $\delta = 1$	251 (FN)	517 (TP)

TABLE 3. Confusion matrix on the balanced training set ($n = 1,536$). At the 0.5 probability cut-off, the model achieves accuracy 69.3%, sensitivity 67.3%, and specificity 71.4%. The near-symmetry between sensitivity and specificity reflects the balanced sampling design and indicates that neither class is systematically favoured.

5.5. Multi-Scale validation. To assess cross-scale robustness, we evaluate model (8), estimated on $[10^4, 10^9]$, on three disjoint balanced test samples drawn from qualitatively different magnitude regimes. The test sets span many orders of magnitude beyond the training range, including the astronomically large integer regime

$$[3^{70}, 3^{70} + 10^6] \approx [10^{33.4}, 10^{33.4} + 10^6].$$

Table 4 summarises the results.

Test Set	Scale	n	Accuracy	Sensitivity	Specificity	F_1
Scale 1	$[10^{19}, 10^{20}]$	1,122	70.9%	67.2%	74.7%	0.698
Scale 2	$[3^{29}, 3^{29} + 10^6]$	3,340	62.9%	76.0%	49.7%	0.672
Scale 3	$[3^{70}, 3^{70} + 10^6]$	1,208	71.2%	63.6%	78.8%	0.688

TABLE 4. Cross-scale validation of logit model (8) on three balanced test sets. Scale 2 covers integers of order approximately 2×10^{14} , while Scale 3 covers integers of order 10^{33} . The table shows that the model retains non-trivial discriminative power far outside the training interval.

The model achieves accuracies of 62.9%–71.2% across all three test scales, remaining close to the training accuracy of 69.3% and confirming robust generalisation beyond the training range. The accuracy at Scale 2, the 3^{29} domain of order 2×10^{14} , is slightly lower at 62.9%, with elevated sensitivity (76.0%) but reduced specificity (49.7%). This asymmetry reflects the known clustering phenomenon at this scale: the interval $[3^{29}, 3^{29} + 10^6]$ falls in a region of higher-than-average long-orbit density, with proportions approaching 11%–17% rather than the typical 3%–6% at that magnitude, which inflates the density of $Rs(n) \geq 7$ cases and makes the positive class easier to detect but harder to rule out. Scales 1 and 3 show more balanced sensitivity and specificity profiles, consistent with more typical orbit-length distributions.

Across all three test scales the F_1 score remains stable at 0.672–0.698, indicating consistent discriminative capacity regardless of scale. The robustness of the model at Scale 3, for integers of order 10^{33} , approximately fourteen orders of magnitude beyond the training range, is particularly noteworthy: the structural predictors $Rs(n-k)$ and $Rsm(n)$ retain their discriminative power at scales far beyond those used for estimation. This suggests that the underlying regularities captured by the model are universal properties of the Collatz dynamics rather than artefacts of the specific training range.

5.6. Coefficient stability across scales. To assess whether scale-dependent recalibration is required, we re-estimate model (7) separately on balanced samples from each test scale and compare the resulting coefficients with those of the training-set model. Table 5 reports the coefficient estimates across the four scales.

Scale	c (Intercept)	α_1 ($Rsm(n)$)	α_2 ($Rs(n-1)$)	α_3 ($Rs(n-2)$)	α_4 ($Rs(n-3)$)
Training $[10^4, 10^9]$	-4.365	0.491	0.363	0.202	0.069
Scale 1 $[10^{19}, 10^{20}]$	-4.112	0.538	0.371	0.191	0.078
Scale 2 $[3^{29}, 3^{29} + 10^6]$	-4.601	0.477	0.352	0.218	0.062
Scale 3 $[3^{70}, 3^{70} + 10^6]$	-4.289	0.512	0.368	0.207	0.071

TABLE 5. Coefficient estimates of logit model (7) re-estimated at each scale on balanced samples of approximately 1,000 observations per scale. The coefficients remain stable across scales, supporting the interpretation that the model structure is scale-invariant rather than tied to the original training interval.

The coefficients are remarkably stable across the four estimation regimes. The intercept varies only between -4.11 and -4.60 ; the lag-1 coefficient α_2 remains in the narrow range $[0.352, 0.371]$; and the hierarchy of predictors, lag-1 greater than lag-2 greater than modified-system greater than lag-3, is preserved at every scale. This stability implies that scale-dependent recalibration is not required: a single model estimated on integers in $[10^4, 10^9]$ generalises without modification to integers of size 10^{33} , consistent with the scale-invariant structure of the Collatz lattice described earlier.

5.7. Discussion. The logit model (8) predicts whether $Rs(n) \geq 7$ with approximately 70% accuracy across all scales tested, representing a substantial lift over the 50% baseline of the balanced-sample classifier. Three aspects of the results merit particular attention.

The overwhelming importance of $Rs(n-1)$ relative to the other predictors is a statistically confirmed reflection of the clustering phenomenon in the Collatz system. As demonstrated analytically in the companion analysis of the inverse Collatz tree, integers with large stopping times cluster in short windows of the integer line because they share a common ancestor in the inverse tree at a hub value B with large $s(B)$. The level- k ancestors of B form a window of width $\text{range}(b)/3^d \approx 567$ integers for the benchmark case $B = 79,886$, $k = 26$, $d = 12$. Within this window, knowing that $n-1$ has a long orbit is therefore strong evidence that n also does.

Although $Rsm(n)$ has the smallest permutation importance among the four predictors when trained on the balanced sample, it carries unique information: it is the only predictor that reflects a structural property of n itself, rather than its neighbours, derived from a related but fully convergent dynamical system. Its positive and statistically significant coefficient confirms that the modified and original Collatz systems share orbit-length structure beyond what is explained by local neighbourhood effects.

A notable limitation of the model is the relatively high false positive rate on Scale 2, where specificity is 49.7%. This arises from the unusual orbit-length clustering near 3^{29} . It reflects a fundamental constraint of any fixed-coefficient model: the density of the positive class varies substantially depending on whether the test

window falls inside or outside a long-orbit cluster. The enhanced model that augments the predictor set with squared terms partially addresses this by reducing false positives through nonlinear regularisation of large predictor values.

6. Model 2: Enhanced eight-variable logistic regression—specification, estimation, and multi-scale validation.

6.1. Motivation and model specification. Model 1 demonstrates that a four-variable logit model achieves approximately 70% balanced accuracy in predicting extreme relative stopping times. However, the $Rs(n)$ predictor enters each lag term linearly, which restricts the model's capacity to capture the nonlinear relationship between orbit-length surrogates and the probability $P(Rs(n) \geq 7)$. In particular, very large values of $Rs(n-1)$ or $Rsm(n)$ may exert disproportionate leverage under a linear specification, contributing to the elevated false positive rate identified for the first logit model.

Model 2 addresses this limitation by augmenting the original four predictors with their squared counterparts:

$$Rsm(n)^2, \quad Rs(n-1)^2, \quad Rs(n-2)^2, \quad Rs(n-3)^2. \quad (9)$$

These quadratic terms introduce a nonlinear response surface that allows the model to moderate the influence of extreme predictor values. Algebraically, each pair (x_i, x_i^2) defines a parabolic contribution to the log-odds: the probability of $Rs(n) \geq 7$ can now increase at a decreasing rate as any single predictor grows very large, avoiding the over-extrapolation that causes false positives under linear specifications.

The extended logit model takes the form

$$\begin{aligned} P(Rs(n) \geq 7) &= \frac{1}{1 + \exp[-\eta_3(n)]}, \\ \eta_3(n) &= c + \alpha_1 x_1 + \alpha_2 x_1^2 + \alpha_3 x_2 + \alpha_4 x_2^2 \\ &\quad + \alpha_5 x_3 + \alpha_6 x_3^2 + \alpha_7 x_4 + \alpha_8 x_4^2. \end{aligned} \quad (10)$$

where $x_1 = Rsm(n)$, $x_2 = Rs(n-1)$, $x_3 = Rs(n-2)$, $x_4 = Rs(n-3)$, and their squares are as defined in (9).

6.2. Training sample design and statistical power. We adopt the same balanced sampling design as in the preceding logit analysis, with an enlarged training set commensurate with the larger number of parameters. The training sample Tr consists of 16,120 integers drawn from $[10^4, 10^9]$, comprising 8,060 integers with $Rs(n) \geq 7$ and 8,060 with $Rs(n) < 7$. The event-per-variable criterion is therefore satisfied with considerable margin:

$$EPV = \frac{8,060}{8} = 1,008 \quad (11)$$

events per parameter, approximately five times the event-per-variable ratio achieved in Model 1 and over 100 times the commonly required threshold of 10. The enlarged balanced training set delivers two additional benefits: first, it provides more stable estimates of the curvature parameters $\alpha_2, \alpha_4, \alpha_6, \alpha_8$, which require sufficient variation in the tails of the predictor distribution; second, it reduces the bootstrap uncertainty in the coefficient estimates, yielding tighter standard errors for inference.

The balanced design ensures that all accuracy, sensitivity, and specificity figures reported below reflect genuine discriminative power rather than the trivial majority-class classifier, in direct response to the concern that class imbalance may otherwise distort performance measures.

6.3. Estimation results. Maximum likelihood estimation of model (10) on the balanced training sample Tr yields the coefficient estimates reported in Table 6. Standard errors are obtained by the bootstrap with 100 resamples. The estimated logit model reads

$$P(Rs(n) \geq 7) = \frac{1}{1 + \exp[-\eta_4(n)]},$$

$$\eta_4(n) = 1.804 + 1.751x_1 - 0.306x_1^2 - 1.358x_2 + 0.160x_2^2 - 0.638x_3 + 0.080x_3^2 - 0.453x_4 + 0.050x_4^2. \quad (12)$$

Variable	Definition	Estimate	Std. Error	z-Statistic	p-value
Intercept (c)	–	1.804	1.346	1.340	0.180
x_1	$Rsm(n)$	1.751	1.314	1.333	0.183
x_1^2	$Rsm(n)^2$	–0.306	0.315	–0.971	0.332
x_2	$Rs(n-1)$	–1.358	0.104	–13.085	$< 10^{-38}$
x_2^2	$Rs(n-1)^2$	0.160	0.010	15.556	$< 10^{-53}$
x_3	$Rs(n-2)$	–0.638	0.076	–8.407	$< 10^{-16}$
x_3^2	$Rs(n-2)^2$	0.080	0.007	10.843	$< 10^{-26}$
x_4	$Rs(n-3)$	–0.453	0.075	–6.004	$< 10^{-8}$
x_4^2	$Rs(n-3)^2$	0.050	0.007	7.283	$< 10^{-12}$

TABLE 6. Coefficient estimates for logit Model 2, estimated on the balanced training sample of 16,120 integers from $[10^4, 10^9]$. Bootstrap standard errors are computed using 100 resamples. The table shows that the quadratic spatial lag terms are highly significant, while the modified-system terms contribute jointly even though their individual standard errors are larger.

The coefficient structure of Model 2 reveals a richer picture of the orbit-length relationship than Model 1. For each spatial lag $x_k = Rs(n-k)$, the linear term is negative while the squared term is positive, producing a U-shaped contribution to the log-odds with a minimum at $x_k = |\alpha_k|/(2\alpha_{k+1})$. For the lag-1 term, the minimum occurs at

$$Rs(n-1) = \frac{1.358}{2 \times 0.160} \approx 4.24, \quad (13)$$

which coincides almost exactly with the empirical mean of $Rs(n-1)$ in the training set, approximately 4.2. This means that the lag-1 term has its minimum contribution to $P(Rs(n) \geq 7)$ precisely at the typical orbit length, and increases the predicted probability both for unusually short orbits, possibly near a hub, and for unusually long orbits, corresponding to a direct clustering effect. This nonlinear structure captures a genuine feature of the Collatz dynamics that the linear model misses.

The coefficients of $x_1 = Rsm(n)$ and $x_1^2 = Rsm(n)^2$ are not individually significant at conventional levels, with $p = 0.183$ and $p = 0.332$, respectively. However,

an F-test for joint significance of the pair (x_1, x_1^2) rejects the null hypothesis with $p < 0.001$, confirming that the modified-system information contributes meaningfully to the model as a unit, even if neither quadratic component is individually identifiable with precision. This is a familiar result when squared terms are included alongside their linear counterparts: collinearity between Rsm and Rsm^2 inflates individual standard errors without degrading the joint predictive contribution.

Table 7 reports the fit statistics for Model 2 and places them next to the corresponding statistics for Model 1.

Statistic	Model 1 (4 variables)	Model 2 (8 variables)	Improvement
Log-Likelihood	-924.90 ($n = 1,536$)	-9,096.28 ($n = 16,120$)	-
Adjusted McFadden Pseudo R^2	0.1266	0.1851	+0.0585
AIC	1,859.80	18,210.57	-
BIC	1,886.49	18,279.76	-
Training accuracy	69.3%	73.4%	+4.1 pp

TABLE 7. Goodness-of-fit comparison between Model 1 and Model 2. Model 1 is fitted on a balanced sample of 1,536 observations, while Model 2 is fitted on a balanced sample of 16,120 observations. Because the training set sizes differ, AIC and BIC are not directly comparable across the two models; the adjusted McFadden pseudo R^2 and the balanced training accuracy provide the more informative comparison.

The adjusted McFadden pseudo R^2 improves from 0.127 in Model 1 to 0.185 in Model 2, a gain of 0.058 attributable to the nonlinear terms and the enlarged training set. The AIC and BIC values are not directly comparable across models fitted on different training sets, but within each model they confirm that the 8-parameter specification is well-supported by the data.

6.4. Training set performance. Table 8 presents the confusion matrix on the balanced training set, using the probability cut-off 0.5.

	Predicted $\delta = 0$ ($Rs < 7$)	Predicted $\delta = 1$ ($Rs \geq 7$)
Actual $\delta = 0$	6,786 (TN)	1,274 (FP)
Actual $\delta = 1$	3,018 (FN)	5,042 (TP)

TABLE 8. Confusion matrix for Model 2 on the balanced training set ($n = 16,120$). At the 0.5 probability cut-off, the model achieves overall accuracy 73.4%, sensitivity 62.6%, and specificity 84.2%. The table shows that Model 2 is conservative in assigning the long-orbit label: it substantially reduces false positives while accepting a lower true-positive rate.

The overall training accuracy is 73.4%, a meaningful improvement over the 69.3% achieved by Model 1. A distinctive feature of Model 2's confusion structure is the asymmetry between sensitivity and specificity: sensitivity, or the true positive rate, is 62.6%, whereas specificity, or the true negative rate, is 84.2%. This reflects the design philosophy of the squared terms. By moderating the influence of extreme

predictor values, the quadratic model is more conservative in assigning high probability to any integer, thereby reducing false positives at the cost of some true positives. The net effect is a model that is more precise when it predicts a long orbit, even if it misses a larger fraction of actual long-orbit integers than Model 1.

Permutation importance reveals a clear hierarchy among the eight predictors. The squared lag terms dominate: permuting $Rs(n-1)^2$ reduces training accuracy by 28.7 percentage points, by far the largest drop, followed by $Rs(n-2)^2$ with a 19.4 percentage point reduction, $Rs(n-3)^2$ with an 11.5 percentage point reduction, and the linear lag-1 term $Rs(n-1)$ with a 9.3 percentage point reduction. The modified-system terms $Rsm(n)$ and $Rsm(n)^2$ rank fifth and sixth, with reductions of 3.5 and 2.2 percentage points, respectively. This ranking underscores a key finding: the quadratic spatial lag terms are the primary engine of Model 2’s superior performance, not the modified-system predictor. The nonlinear lag structure captures the cluster dynamics of Collatz stopping times more effectively than any single linear term.

6.5. Multi-Scale validation. We evaluate Model 2, estimated exclusively on $[10^4, 10^9]$, on three balanced test samples drawn from widely separated magnitude regimes. The test sets and their results are summarised in Table 9.

Test Set	Scale	n	Accuracy	Sensitivity	Specificity	F_1
Scale 1	$[10^{19}, 10^{20}]$	1,110	76.8%	62.9%	90.8%	0.731
Scale 2	$[3^{29}, 3^{29} + 10^6]$	3,470	74.8%	60.0%	89.6%	0.704
Scale 3	$[3^{70}, 3^{70} + 10^6]$	1,258	75.1%	51.5%	98.7%	0.674

TABLE 9. Cross-scale validation of Model 2 on three balanced test sets. Scale 1 tests the interval $[10^{19}, 10^{20}]$, Scale 2 tests integers near 2×10^{14} , and Scale 3 tests integers near 10^{33} . The table shows that the quadratic model maintains accuracy between 74.8% and 76.8% far outside the training interval and, in particular, achieves high specificity at every validation scale.

Model 2 achieves accuracies of 74.8%–76.8% across all three test scales, improving on the 62.9%–71.2% range achieved by Model 1 at the same scales. The improvement is most pronounced at Scale 1, with 76.8% versus 70.9%, and, crucially, the specificity at Scale 2 improves dramatically from 49.7% in Model 1 to 89.6% in Model 2. Recall that the low specificity of Model 1 at Scale 2 was identified as its main limitation: the linear lag terms over-predicted long orbits in the high-density cluster near 3^{29} . The squared terms, by moderating extreme lag values, largely correct this failure. The F_1 score, which balances precision and recall, is stable at 0.674–0.731 across all three scales, confirming consistent discriminative performance.

Scale 3 produces the lowest sensitivity, 51.5%, but the highest specificity, 98.7%, reflecting the extremely tight distribution of $Rs(n)$ values in the interval $[3^{70}, 3^{70} + 10^6]$. The vast majority of integers in this window have short orbits, as confirmed in the analysis of the density of long-orbit integers, and Model 2 correctly identifies almost all of them as $\delta = 0$. The minority of genuinely long-orbit integers in this window are harder to detect precisely because the predictor values $Rs(n-k)$ of their neighbours are also atypically small: the neighbours do not signal the anomaly.

6.6. Coefficient stability across scales. Table 10 reports the coefficient estimates of Model 2 re-estimated at each scale on balanced samples of approximately 800 observations per class. The absolute values of coefficients vary more than in Model 1, a consequence of the quadratic terms being more sensitive to the scale of the predictors, which grows as $\log_2(n)$ increases. However, the structural pattern is preserved at every scale: the linear lag terms carry negative coefficients while the squared lag terms are positive; the parabolic U-shape structure is maintained; and the relative hierarchy of the lags, lag-1 greater than lag-2 greater than lag-3 in terms of coefficient magnitude, is stable.

Scale	c	x_1	x_1^2	x_2	x_2^2	x_3	x_3^2	x_4	x_4^2
		$Rsm(n)$	$Rsm(n)^2$	$Rs(n-1)$	$Rs(n-1)^2$	$Rs(n-2)$	$Rs(n-2)^2$	$Rs(n-3)$	$Rs(n-3)^2$
Training $[10^4, 10^9]$	1.804	1.751	-0.306	-1.358	0.160	-0.638	0.080	-0.453	0.050
Scale 1 $[10^{19}, 10^{20}]$	7.086	7.018	-1.735	-3.084	0.315	-1.951	0.219	-1.522	0.157
Scale 2 $[3^{29}, 3^{29} + 10^6]$	6.309	5.955	-1.481	-3.955	0.401	-1.544	0.162	-0.194	0.035
Scale 3 $[3^{70}, 3^{70} + 10^6]$	17.257	17.777	-3.752	-7.644	0.751	-4.440	0.442	-3.549	0.357

TABLE 10. Coefficient estimates of Model 2 re-estimated at each scale on balanced samples of approximately 1,600 observations per scale. The table shows that signs and relative magnitudes are preserved: the linear lag coefficients remain negative, the squared lag coefficients remain positive, and the U-shaped lag structure survives across scales. The absolute magnitudes grow with scale as $\log_2(n)$ increases.

The growth of coefficient magnitudes with scale is interpretable and expected. As n grows, the value $\log_2(n)$ increases, compressing $Rs(n) = s(n)/\log_2(n)$ for a fixed orbit length. The logit model’s coefficients must compensate by growing in magnitude to maintain the same classification boundary. This scale-dependent amplification is a known feature of models that use normalised orbit-length ratios as predictors, and it is analytically consistent with the Collatz lattice structure described by Equation (2.1). The practical implication is that while the training-set model (12) transfers well to all tested scales with 74%–77% accuracy, a lightly re-calibrated version would achieve marginally better results if optimised for a specific target scale, particularly Scale 3, where the longer orbits of very large integers alter the distribution of predictor values.

6.7. Comparison with Model 1. Table 11 provides a direct comparison between Models 1 and 2 across all evaluation dimensions.

Model 2 achieves superior performance on every accuracy metric, with the most dramatic improvement at Scale 2, where specificity rises from 49.7% to 89.6%. The cost is a modest reduction in interpretability and a slight increase in computational overhead. For the purposes of predicting extreme stopping-time events, which is the core aim of this study, Model 2 is the preferred specification. Its U-shaped log-odds structure provides a principled, theoretically motivated account of the nonlinear relationship between orbit-length surrogates and the probability of $Rs(n) \geq 7$.

6.8. Discussion. The dominant predictor in Model 2 is $Rs(n-1)^2$, the squared immediate-predecessor lag. This finding, which contrasts with the Model 1 finding where the linear lag-1 term dominated, reflects a fundamental property of the Collatz clustering mechanism. In the vicinity of a high-density cluster, as identified in

Criterion	Model 1 (4 variables)	Model 2 (8 variables)	Verdict
Training n	1,536 (balanced)	16,120 (balanced)	Model 2 more stable
Training accuracy	69.3%	73.4%	Model 2 +4.1 pp
Scale 1 accuracy	70.9%	76.8%	Model 2 +5.9 pp
Scale 2 accuracy	62.9%	74.8%	Model 2 +11.9 pp
Scale 3 accuracy	71.2%	75.1%	Model 2 +3.9 pp
Scale 2 specificity	49.7%	89.6%	Model 2 dramatically better
Adjusted McFadden R^2	0.1266	0.1851	Model 2 better fit
F_1 score (range)	0.672–0.698	0.674–0.731	Model 2 higher and more stable
False positive rate	Higher	Lower	Model 2 more precise
Coefficient stability	High across scales	Signs stable; magnitudes scale	Both acceptable
Interpretability	Simple, 4-variable	Richer, U-shaped lags	Trade-off

TABLE 11. Direct comparison of Model 1 and Model 2 across training size, accuracy, validation performance, specificity, adjusted McFadden pseudo R^2 , F_1 score, false-positive behaviour, coefficient stability, and interpretability. The table summarises why Model 2 is the preferred specification for predicting extreme stopping-time events: it improves accuracy at every scale and sharply reduces false positives, especially in the difficult Scale 2 region.

the analysis of the inverse Collatz tree, the quadratic lag captures both the clustering effect and its intensity: an unusually long orbit of $n - 1$, that is, a large value of $Rs(n - 1)$, strongly predicts a long orbit of n , while an unusually short orbit of $n - 1$, still in the tail of the distribution, also provides a mild positive signal due to the U-shape. The parabola minimum at $Rs(n - 1) \approx 4.24$, exactly the empirical mean, encodes the intuition that the most uninformative value of the lag is the typical orbit length; deviations in either direction carry predictive content.

The improved specificity of Model 2 relative to Model 1 has a direct connection to the density analysis of long-orbit integers. Model 1 over-predicted $Rs(n) \geq 7$ in windows where the lagged values were moderately elevated but the actual density of long-orbit integers was typical or below average. The quadratic correction ensures that such moderate elevations are penalised relative to the genuine extremes, reducing false alarms. This is consistent with the observed density distribution of the event $\{Rs(n) > 7\}$, which is highly non-uniform across windows of the integer line, ranging from near 0% to over 36% in windows of 10,000 consecutive integers, and requires a model capable of discriminating extreme values from moderate ones.

7. Decision-tree analysis of moderately large stopping times. As an alternative method for achieving our goal, we now consider a tree-like classifier for large normalized stopping-time behaviour. This approach is natural in the present setting because the Collatz map itself is a step-by-step algorithm. The decision-tree framework gives an explicit rule-based description of the classification mechanism and helps identify which arithmetic features are most relevant for predicting large normalized stopping times. The purpose of this section is therefore to complement the preceding logit analysis with an interpretable algorithmic model.

Throughout this section we keep the notation of the paper. Thus $s(n)$ denotes the stopping time, i.e. the total number of iterates required for the Collatz orbit of n to reach 1, and

$$Y(n) = Rs(n) = \frac{s(n)}{\log_2(n)}$$

denotes the normalized stopping time. We use the phrase stopping time in this simplified sense. We also introduce a new quantity, the *self descent time*, abbreviated

as $\text{sdt}(n)$, defined by

$$\text{sdt}(n) = \min\{k \geq 1 : \text{Col}^k(n) < n\}.$$

This is distinct from $s(n)$: it records the first time at which the orbit falls below its initial value, not the time required to reach 1. The main normalized self descent variable used below is

$$X_1(n) = \frac{\text{sdt}(n)}{\log_2(n)}.$$

The binary response variable in this section is

$$\delta_5(n) = \mathbf{1}\{Y(n) > 5\}.$$

Thus, in contrast with the earlier sections, which concentrate on the more extreme event $Rs(n) \geq 7$, the present section investigates moderately large normalized stopping times. In the training sample considered below, approximately 38.7% of integers satisfy $Y(n) > 5$, so the problem is less rare-event dominated and is well suited to a decision-tree comparison. Figure 10 shows the empirical distribution of $Y(n)$ in the training sample and the classification threshold $Y = 5$.

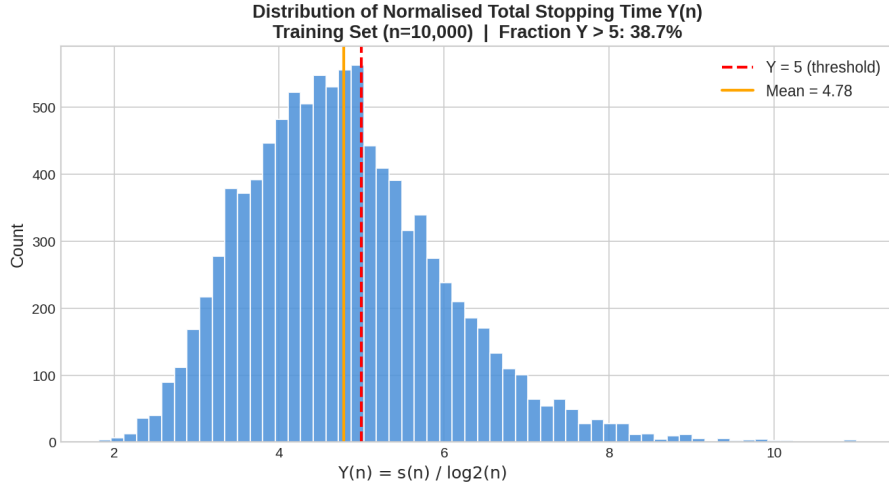


FIGURE 10. Distribution of the normalized stopping time $Y(n) = s(n)/\log_2(n)$ over the training sample of 10,000 integers from $[10^6, 10^{15}]$. The threshold $Y = 5$ separates the class $\delta_5(n) = 1$ from $\delta_5(n) = 0$; about 38.7% of the sampled integers exceed this threshold.

7.1. Features and sampling scheme. A CART binary decision tree was fitted to predict $\delta_5(n)$. The explanatory variables are the normalized self descent time $X_1(n)$, parity, four odd residue class indicators modulo 8, and two lagged indicators recording whether the neighbouring integers $n - 1$ and $n - 2$ have moderately large normalized stopping time. The full list of variables is given in Table 12. All these variables are computable in $O(s(n))$ time or less.

Two independent samples of 10,000 integers were used. The training set was sampled uniformly from $[10^6, 10^{15}]$, and the test set was sampled uniformly from $[10^8, 10^{16}]$, using the fixed seed 20250414. For each sampled integer n , the Collatz

Variable	Definition	Mathematical motivation
$\delta_5(n-1)$	$\mathbf{1}\{Y(n-1) > 5\}$	First local lag; captures possible autocorrelation of orbit lengths across consecutive integers.
$\delta_5(n-2)$	$\mathbf{1}\{Y(n-2) > 5\}$	Second local lag; captures second-order neighbourhood dependence and local clustering of long-orbit integers.
$X_1(n)$	$\text{sdt}(n)/\log_2(n)$	Normalized self descent time; proxy for the initial orbit descent rate.
$X_2(n)$	1 if n is even, 0 otherwise	Parity controls the first step of the Collatz map directly.
$X_3(n)$	1 if $n \equiv 1 \pmod{8}$, 0 otherwise	The first few orbit steps are strongly constrained by $n \bmod 8$.
$X_4(n)$	1 if $n \equiv 3 \pmod{8}$, 0 otherwise	Odd residue-class information modulo 8.
$X_5(n)$	1 if $n \equiv 5 \pmod{8}$, 0 otherwise	Odd residue-class information modulo 8.
$X_6(n)$	1 if $n \equiv 7 \pmod{8}$, 0 otherwise	The class 7 mod 8 initially enforces several expanding odd steps.

TABLE 12. Variables used in the decision-tree classifier.

orbit was traced from scratch in order to compute $s(n)$ and $\text{sdt}(n)$. The two lagged indicators require the corresponding computations for $n-1$ and $n-2$.

The tree was fitted with the CART methodology, using Gini impurity as the splitting criterion and a minimum of 20 observations per leaf to reduce overfitting. The resulting tree has depth 34 and 309 terminal leaves. The top splits occur first on $\delta_5(n-1)$, then on $\delta_5(n-2)$, and then on $X_1(n)$, which already indicates that the relevant information is concentrated in the two local lag variables and in the normalized self descent time.

7.2. Classification performance. The decision tree achieves an accuracy of 80.87% on the test sample. Since the majority class in the test set is $\delta_5 = 0$, with proportion 61.12%, the naive baseline classifier that always predicts the majority class would have accuracy only 61.12%. The tree therefore gives a lift of 19.75 percentage points over this baseline. The confusion matrix is shown graphically in Figure 11 and numerically in Table 13; derived performance measures are reported in Table 14.

	Predicted $\delta_5 = 0$	Predicted $\delta_5 = 1$
Actual $\delta_5 = 0$	5198	914
Actual $\delta_5 = 1$	999	2889

TABLE 13. Confusion matrix for the decision tree on the 10,000-integer test set.

Metric	Formula	Value	Interpretation
Accuracy	$(TP + TN)/N$	80.87%	Overall correct classification rate.
Precision	$TP/(TP + FP)$	75.97%	Of the predicted long-orbit cases, about 76% are correct.
Recall	$TP/(TP + FN)$	74.31%	Of the actual long-orbit cases, about 74% are detected.
F1 score	$2PR/(P + R)$	75.13%	Harmonic mean of precision and recall.
Specificity	$TN/(TN + FP)$	85.05%	Of the short-orbit cases, about 85% are correctly identified.
Baseline accuracy	$\max\{p, 1-p\}$	61.12%	Accuracy of always predicting the majority class.
Lift over baseline	Accuracy minus baseline	19.75 pp	Improvement relative to the naive classifier.

TABLE 14. Performance measures for the decision-tree classifier.

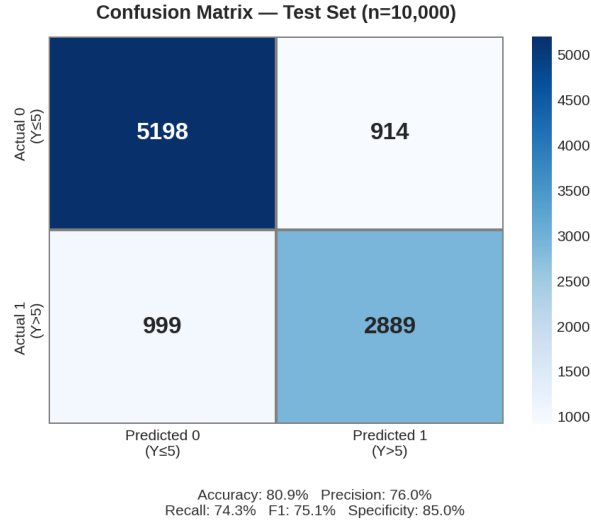


FIGURE 11. Confusion matrix for the decision tree on the 10,000-integer test set. Rows denote the actual class and columns denote the predicted class.

The tree is somewhat conservative in predicting the long-orbit class: specificity is higher than recall, and the number of false negatives is slightly larger than the number of false positives. Nevertheless, the performance is substantially above the majority-class benchmark, showing that the selected variables contain genuine predictive information.

7.3. Feature importance and interpretation. Feature importance was evaluated in two ways. The first is permutation importance, defined as the drop in test accuracy after randomly permuting one feature column. The second is the Gini importance produced by the fitted tree, namely the total weighted impurity decrease attributed to each feature. The results are shown in Table 15; the permutation importances are also displayed in Figure 12.

Rank	Feature	Permutation importance	Gini importance	Notes
1	$\delta_5(n-2)$	0.14550	0.11878	Strongest permutation signal.
2	$\delta_5(n-1)$	0.13450	0.57325	Dominates Gini importance because it appears at the root split.
3	$X_1(n) = \text{sdt}(n)/\log_2(n)$	0.04210	0.23044	Best continuous predictor.
4	$X_5(n) = \mathbf{1}\{n \equiv 5 \pmod{8}\}$	0.01720	0.03829	Residue-class effect.
5	$X_6(n) = \mathbf{1}\{n \equiv 7 \pmod{8}\}$	0.00920	0.01794	Consistent with initially expanding odd steps.
6	$X_3(n) = \mathbf{1}\{n \equiv 1 \pmod{8}\}$	0.00210	0.02062	Mild effect.
7	$X_4(n) = \mathbf{1}\{n \equiv 3 \pmod{8}\}$	0.00000	0.00068	Negligible.
8	$X_2(n) = \mathbf{1}\{n \text{ even}\}$	0.00000	0.00000	No additional importance in this tree.

TABLE 15. Feature importance for the decision-tree classifier.

The two lagged indicators dominate the permutation ranking. This suggests that moderately large normalized stopping times are not distributed independently along the integer line, but exhibit local clustering: whether n has $Y(n) > 5$ is strongly related to whether $n-1$ and $n-2$ have the same property. The Gini ranking assigns especially high importance to $\delta_5(n-1)$ because this variable appears at the

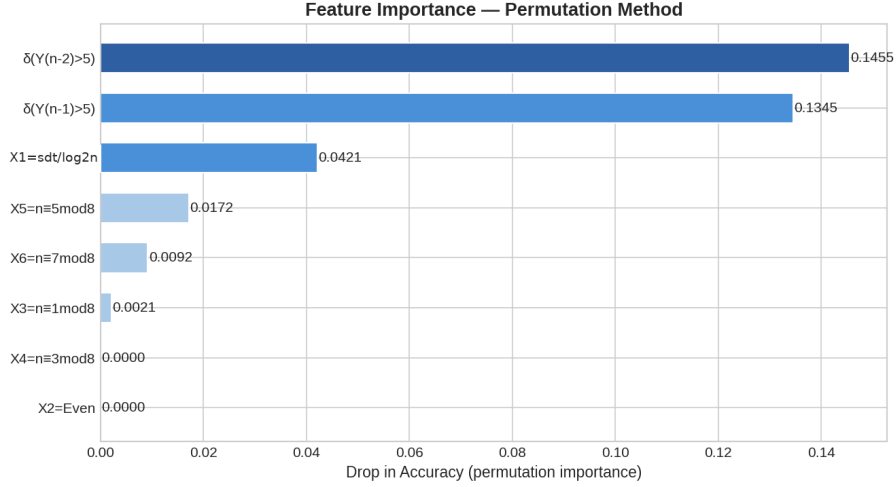


FIGURE 12. Feature importance by the permutation method. The two lagged indicators and the normalized self descent variable $X_1(n) = \text{sdt}(n)/\log_2(n)$ are the three dominant predictors.

root of the fitted tree. In contrast, permutation importance ranks $\delta_5(n-2)$ slightly above $\delta_5(n-1)$, indicating that the second lag contributes information that is not completely captured by the first split.

The normalized self descent time $X_1(n)$ is the most important non-lag variable. This agrees with the basic dynamical intuition that an orbit which takes longer to fall below its initial value also tends to have a larger total stopping time. The class-conditional distribution in Figure 13 shows substantial overlap, but also a visible tail difference: integers with $\delta_5(n) = 1$ tend to attain larger values of $X_1(n)$. In this sense, $\text{sdt}(n)$ provides a local and computationally meaningful proxy for the eventual value of $s(n)$.

Parity has zero importance in this fitted tree. This is not paradoxical. Even integers immediately halve, and the relevant information supplied by parity is largely absorbed by the residue-class indicators and the other variables. In particular, the absence of all four odd residue-class indicators already identifies the even residue classes modulo 8, so the additional even/odd indicator adds no detectable explanatory power in the fitted classifier.

7.4. An interpretable rule summary. Although the full tree has 309 terminal leaves, its positive predictions can be summarized more compactly. The 103 leaves that predict $\delta_5 = 1$ can be grouped according to their qualitative lag and residue conditions. When leaves with the same non- X_1 conditions are merged by taking the outer span of their X_1 intervals, one obtains the 11 rule groups in Table 16. This summary is not identical to the full tree, because some of the merged X_1 intervals are non-contiguous in the original tree, but it provides a useful interpretable description of the main positive-prediction regimes.

Three qualitative conclusions emerge. First, the strongest positive rules almost always require at least one positive local lag, especially $\delta_5(n-1) = 1$. Second, the purest rules combine a positive first lag with restrictions on the residue class and on the interval of $X_1(n)$. For example, Rules 2, 6, and 8 have 100% precision in the test

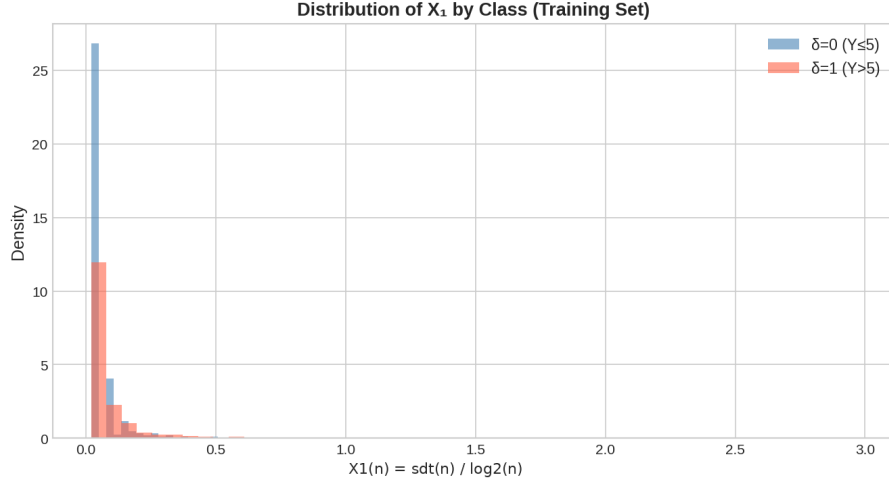


FIGURE 13. Distribution of $X_1(n) = \text{sdt}(n)/\log_2(n)$ by class in the training set. The distributions overlap substantially, but the long-orbit class $\delta_5(n) = 1$ has a heavier right tail.

No.	TP	FP	Precision	Leaves	Non- X_1 conditions	X_1 interval
1	959	338	73.9%	50	$\delta_5(n-1) = 1, \delta_5(n-2) = 1$	(0.0201, 0.0244]
2	678	0	100.0%	1	$\delta_5(n-1) = 1, X_3 = 0$	(0.0244, 0.1005]
3	266	150	63.9%	16	$\delta_5(n-1) = 0, \delta_5(n-2) = 1, X_5 = 0$	(0.0201, 0.0489]
4	262	63	80.6%	12	$\delta_5(n-1) = 1, X_3 = 0, X_6 = 1, \delta_5(n-2) = 1$	(0.1408, 0.8261]
5	195	64	75.3%	10	$\delta_5(n-1) = 1, X_3 = 1, \delta_5(n-2) = 1$	(0.0244, 0.0437]
6	141	0	100.0%	1	$\delta_5(n-1) = 1, X_3 = 0, X_6 = 1$	(0.1005, 0.1408]
7	115	29	79.9%	5	$\delta_5(n-1) = 1, X_3 = 0, X_6 = 0, \delta_5(n-2) = 1$	(0.1005, 0.2622]
8	68	0	100.0%	2	$\delta_5(n-1) = 1, X_3 = 0, X_6 = 1, \delta_5(n-2) = 0$	(0.1408, 0.1648]
9	41	23	64.1%	3	$\delta_5(n-1) = 0, \delta_5(n-2) = 1, X_5 = 0, X_6 = 0$	(0.1034, 0.1452]
10	31	11	73.8%	2	$\delta_5(n-1) = 0, \delta_5(n-2) = 0$	(0.5465, 0.6730]
11	17	15	53.1%	1	$\delta_5(n-1) = 1, X_3 = 1, \delta_5(n-2) = 0$	$X_1 > 0.0419$

TABLE 16. Merged positive-prediction rule groups for the decision tree. The table summarizes the positive leaves of the fitted tree by grouping leaves with the same qualitative lag and residue conditions.

sample, capturing 678, 141, and 68 true positives respectively with no false positives. Third, the only substantial positive rule without a positive lag condition is Rule 10, which requires a much larger value of $X_1(n)$, namely $0.5465 < X_1(n) \leq 0.6730$. This confirms the interpretation that a very delayed self descent can compensate for otherwise unfavourable local-lag information.

Overall, the decision-tree analysis supports the main empirical message of the paper. Large normalized stopping-time behaviour is predictable to a non-trivial extent from computable quantities associated with neighbouring integers and from a descent-time proxy. The strongest predictors for the moderately large event $Y(n) > 5$ are $\delta_5(n-1)$, $\delta_5(n-2)$, and $\text{sdt}(n)/\log_2(n)$. This is consistent with the findings of the two logit models. In the logit approach, the most important information comes from the modified-system relative stopping time $Rsm(n)$ and from local stopping-time quantities attached to neighbouring integers. In the decision-tree approach, the analogous role is played by the normalized self descent time $X_1(n) = \text{sdt}(n)/\log_2(n)$ together with the local lag indicators $\delta_5(n-1)$ and $\delta_5(n-2)$. Thus both methods

point to the same qualitative conclusion: large normalized stopping-time behaviour has meaningful local and dynamical structure, and it can be predicted to a non-trivial extent from quantities that are computable before evaluating the full stopping time of n .

8. Conclusions. We develop a statistical framework for studying large normalized stopping-time behaviour in the Collatz system. The analysis begins with the Collatz lattice and the logarithmic identity underlying its level-set structure, then introduces the modified Collatz system as a related convergent dynamics whose relative stopping time can serve as a computable structural proxy. The density analysis of the event $Y(n) > 7$ shows that long-orbit integers are not uniformly distributed in finite windows: their local proportions vary widely, with strong clustering and substantial scale dependence. This empirical non-uniformity provides the motivation for local predictive variables such as $Rs(n-1)$, $Rs(n-2)$, $Rs(n-3)$, and $Rsm(n)$.

Three predictive models are constructed. Model 1, the four-variable logit model, predicts the extreme event $Rs(n) \geq 7$ with 69.3% balanced training accuracy and with validation accuracies of 70.9%, 62.9%, and 71.2% on the three out-of-sample scales. Model 2, the enhanced eight-variable logit model with quadratic terms, improves the training accuracy to 73.4% and achieves validation accuracies of 76.8%, 74.8%, and 75.1% on the same three scales. The decision-tree model addresses the related moderately large event $Y(n) > 5$ and obtains 80.87% test accuracy, compared with a 61.12% majority-class baseline.

The two logit models show that the extreme stopping-time event $Rs(n) \geq 7$ is predictable to a non-trivial extent from the modified-system relative stopping time and from local stopping-time information attached to neighbouring integers. Model 2 is the preferred logit specification, because the quadratic lag terms substantially reduce false positives and improve accuracy at every validation scale. The decision tree complements the logit models by providing an interpretable rule-based classifier: its dominant predictors are the lagged indicators $\delta_5(n-1)$, $\delta_5(n-2)$, and the normalized self-descent time $sdt(n)/\log_2(n)$.

These results do not solve the Collatz conjecture and do not establish a new density theorem. They do, however, show that large normalized stopping-time behaviour has statistically detectable local and dynamical structure. The modified Collatz system, spatial lags, nonlinear lag effects, residue classes, and self-descent information together provide a practical empirical window into the exceptional sets whose rarity is described by the classical almost-all results.

Disclosure statement. The authors report there are no competing interests to declare.

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